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Project Title:

AetherRat: C2 Agent Development & Offensive Tooling Operations in Kali Linux

Problem Statement:

Red team operators must be proficient in both utilizing industry-standard offensive security distributions like Kali Linux and building custom implants. Off-the-shelf tools are frequently flagged by modern EDRs and Antivirus solutions. This project addresses the need for custom, lightweight Remote Access Modules while integrating established Kali Linux security tools for target reconnaissance, payload delivery, and egress traffic inspection.

Objectives:

- To develop a custom Python-based C2 server and agent capable of remote command execution and system telemetry retrieval.
- To master key Kali Linux tools (Nmap, MSFvenom, Wireshark, Apache) during target reconnaissance, payload packaging, and network traffic analysis.
- To secure C2 communication using AES/XOR encryption to prevent plain-text packet inspection by network defenders.
- To document blue-team detection strategies and packet-level signatures for defensive monitoring.

Proposed Solution / Methodology:

1. **Reconnaissance & Setup:** Use **Nmap** on Kali Linux to scan target lab VMs and map open services.
2. **C2 & Agent Development:** Build a Python C2 listener on Kali Linux and a client agent on the target machine for command execution (`cmd`/`powershell`) and file transfer.
3. **Payload Packaging & Delivery:** Utilize **MSFvenom / PyInstaller** on Kali to package the Python agent into a standalone executable, hosting it via a local **Apache / Python HTTP server** for simulated delivery.
4. **Traffic Analysis & Defense:** Capture and analyze beaconing traffic using **Wireshark** to verify encryption and formulate detection rules.

Technologies & Tools to be Used:

- **OS & Platform:** Kali Linux (Attacker Host), Windows 10/11 VM (Target Host)
- **Kali Tools:** Nmap, Wireshark, MSFvenom, Apache2, Netcat
- **Programming Language:** Python 3 (`socket`, `subprocess`, `cryptography`, `os`)

Scope of the Project:

This project is strictly confined to an isolated Virtual Machine lab environment running Kali Linux and Windows. Testing on unauthorized networks, mass phishing, and destructive malware deployment are strictly out of scope.

Expected Outcomes:

- Practical hands-on expertise in core Kali Linux tools (Nmap, Wireshark, MSFvenom, Web Servers).
- A functional Python C2 server and agent operating over encrypted communication channels.
- A complete Red Team workflow report detailing reconnaissance, delivery, execution, and network detection signatures.

Timeline / Work Plan (1-Week Schedule):

- **Days 1–2:** Perform network reconnaissance using **Nmap** on Kali Linux and set up the Python C2 listener architecture.
- **Days 3–4:** Write the client agent logic for remote execution (`cmd`/`powershell`) and integrate AES/XOR encryption.
- **Days 5–6:** Package the agent executable using Kali Linux tools (**PyInstaller/MSFvenom**), host delivery via Apache, and capture traffic in **Wireshark**.
- **Day 7:** Finalize lab testing, compile blue-team detection rules, and complete the project report and presentation.